

PINN 3D

$$\nabla^2 \phi(\mathbf{r}) = \kappa^2 \phi(\mathbf{r}) - \frac{\rho_f(\mathbf{r})}{\epsilon}$$

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Model      : Vanilla 3D LPB-PINN (no Fourier, no DD)
Parameters: 12,801
PDE        : -∇²u + κ²u = ρ on sphere r ≤ 1.0
κ          : 1.0 (Debye screening length inverse)
500 | 5.01e-04 | 0.0010 | 2.03e-03 | 9.93e-10 | 75.8 | 39.7s
1000 | 6.34e-06 | 0.0001 | 2.67e-04 | 1.21e-10 | 74.9 | 77.2s
1500 | 3.33e-07 | 0.0000 | 4.23e-05 | 1.98e-11 | 76.3 | 115.3s
2000 | 5.26e-07 | 0.0000 | 7.12e-05 | 2.39e-11 | 76.2 | 153.4s
2500 | 1.29e-06 | 0.0000 | 8.26e-05 | 3.25e-11 | 75.1 | 191.0s
3000 | 5.66e-07 | 0.0000 | 7.55e-05 | 2.11e-11 | 78.1 | 230.0s
3500 | 2.00e-06 | 0.0001 | 1.76e-04 | 7.13e-11 | 127.4 | 293.7s
4000 | 5.41e-06 | 0.0001 | 2.73e-04 | 1.04e-10 | 138.2 | 362.8s
4500 | 6.49e-07 | 0.0000 | 8.35e-05 | 4.93e-11 | 137.7 | 431.7s
5000 | 3.67e-07 | 0.0000 | 4.69e-05 | 1.78e-11 | 137.9 | 500.7s
5500 | 1.88e-07 | 0.0000 | 4.49e-05 | 2.38e-11 | 136.0 | 568.7s
6000 | 2.37e-07 | 0.0000 | 3.61e-05 | 1.63e-11 | 136.2 | 636.8s
6500 | 1.13e-06 | 0.0001 | 1.15e-04 | 4.53e-11 | 138.0 | 705.8s
7000 | 8.05e-08 | 0.0000 | 3.08e-05 | 1.58e-11 | 124.8 | 768.2s
7500 | 5.50e-08 | 0.0000 | 1.78e-05 | 6.45e-12 | 163.0 | 849.7s
8000 | 4.90e-08 | 0.0000 | 1.08e-05 | 7.05e-12 | 165.4 | 932.4s
8500 | 4.15e-08 | 0.0000 | 8.70e-06 | 4.75e-12 | 166.8 | 1015.8s
9000 | 3.99e-08 | 0.0000 | 6.68e-06 | 4.95e-12 | 82.8 | 1057.2s
9500 | 3.91e-08 | 0.0000 | 6.91e-06 | 5.23e-12 | 85.6 | 1100.0s
10000 | 3.93e-08 | 0.0000 | 7.99e-06 | 5.97e-12 | 82.2 | 1141.2s

Total training time : 1141.2s (19.02 mins)
Avg time per epoch : 113.9 ms
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Final L2 Relative Error : 0.000003
Final Max Absolute Error : 8.046627e-06
Final BC Violation      : 5.712764e-12 + machine precision
Total Training Time     : 1141.2s
Avg Time per Epoch      : 113.91 ms
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Vanilla 3D LPB-PINN Baseline — Total: 1141s | Avg: 113.9ms/epoch
 $-\nabla^2 u + \kappa^2 u = \rho$, $\kappa=1.0$, $R=1.0$ | No Fourier Features | No DD

